



*Let the mind manage the body
Que l'esprit gère le corps*

**MAURITIUS
EXAMINATIONS
SYNDICATE**

**NATIONAL
CERTIFICATE
OF EDUCATION**

MATHEMATICS

Specimen paper
for first assessment in October 2020

Acknowledgments

The MES would like to place on record its gratitude and appreciation to all those who contributed to the development of the specimen assessment materials - Educators (from the Regular and Extended programmes), academics from the MIE , the University of Mauritius, the Open University and the University of Technology Mauritius, representatives of the Ministry of Education and Human Resources, Tertiary Education and Scientific Research and representatives of recognised unions - who, at different stages in the development of the assessment, have been members of Technical Committees, Validation Committees and Secondary School Examination Committees that were set up by the MES. We are also grateful to the Rectors, Educators and students who took part in the trialling process of the specimen papers. The contribution of all these stakeholders provided us with vital information and feedback which fed into the production of the specimen papers.

1. Background

At the end of the Nine-Year Continuous Basic Education (NYCBE) cycle, all students from the Regular and Extended programmes take the National Certificate of Education (NCE) Assessment. This assessment is in line with the philosophy defined in the National Curriculum Framework (NCF) Grades 7, 8 and 9 (MIE, 2016) and the learning outcomes (LOs) detailed in the NCF Syllabus Grades 7, 8 & 9 (TLS) (MIE, 2019).

The assessment will be carried out in the following subjects:

- English
- Mathematics
- French
- Science
- Information and Communication Technology (ICT)
- Technology Studies
- Business and Entrepreneurship Education (BEE)
- Social and Modern Studies (SMS)
- Art and Design
- An optional core subject (Asian Languages, Arabic and Kreol Morisien, if chosen by the candidate)

A 7-point Grading structure will be used in each subject, as illustrated in **Table 1** below.

Numerical Grade	Marks
1	85 and above
2	75 and above but below 85
3	65 and above but below 75
4	55 and above but below 65
5	45 and above but below 55
6	35 and above but below 45
7	Less than 35

Table 1: Grading Structure for the NCE Assessment

2. Purpose of the NCE Assessment

The main purpose of the NCE Assessment is to measure and certify learning that has taken place at the end of the NYCBE cycle. The information gathered from the assessment will be used for:

- **Certification**

Meeting the minimum requirements on the NCE Assessment (see the Award Rules in the *Annual Programme for the National Certificate of Education (NCE) Assessment for 2020*) will lead to the candidate being conferred an NCE certificate which will be recognised at Level 2 on the National Qualifications Framework (NQF).

- **Promotion to Grade 10**

Assessment results from the NCE will guide schools in determining whether students get promoted to Grade 10.

- **Orientation**

The NCE assessment will provide information to guide students as to whether they want to continue in the general or in the technical/vocational stream. Within the general stream, it may guide students in their choice of subjects as from Grade 10.

- **Admission to Academies.**

Performance in the NCE Assessment will determine whether candidates are admitted to academies.

The following extract from the Education Act indicates the criteria for admission to academies:

Priority of admission to Grade 10 in an Academy shall be determined on the basis of the grade aggregate and the relative performance of the eligible pupil in the best 8 core subjects, including English, French and Mathematics, at the NCE assessment and the choice of the responsible party ...

3. Guiding principles in Assessment

A number of key principles of assessment guided the development of the NCE Assessment (Standards for Educational and Psychological Testing, 2014).

3.1 Validity

Validity is a central concept in assessment. In simple terms, it refers to the extent to which an assessment accurately measures what it is supposed to be measuring. Validity also refers to the extent to which the assessment is providing evidence of candidates' achievement levels.

3.2 Reliability

Reliability, another crucial concept in assessment, refers to producing reliable, stable and consistent results over time. Ensuring reliability requires clear and consistent processes for the setting, marking and grading of the NCE Assessment.

3.3 Impact

The NCE Assessment aims at having positive effects on teaching and learning with positive washback into the curriculum and into the educational system. An important consideration during the development of this assessment was the potential impact that it would have on the life chances of candidates, allowing for maximum inclusion and retention of students in the system while maintaining standards.

3.4 Fairness

Needs and characteristics of learners were considered in the design of the NCE Assessment so as not to disadvantage any group or individual. Care has been taken to minimise cultural and gender biases and to accommodate the different abilities and the social, cultural and linguistic backgrounds of candidates.

4. Paper Design

Three Assessment Objectives (AOs), defined in the *Annual Programme for the National Certificate of Education (NCE) Assessment 2020*, underpin the design of the Mathematics paper:

- **AO1 - Knowing**

Questions assessing *knowing* evaluate recall and use of facts, concepts, rules and procedures which learners need to solve problems. In general, these require the candidate to carry out standard mathematical procedures. These questions may be with or without a context.

- **AO2 - Applying**

Questions assessing *applying* focus on learners' ability to apply their mathematical knowledge and skills to solve routine problems. In these questions, the candidate is required to think of an appropriate method that can be used in a given context.

- **AO3 - Reasoning**

In general, questions assessing *reasoning* require the candidate to deal with routine and non-routine problems which may be multi-step and may be set in complex contexts. Candidates will be expected to demonstrate their ability to devise strategies to solve mathematical problems. Such strategies may encompass more than one method that will lead to a correct solution. The context of these questions may be unfamiliar to the candidate.

Questions assessing *Knowing*, *Applying* and *Reasoning* may require low, average or high order thinking skills and, thus, can vary in terms of difficulty level.

The skills and competencies which will be assessed under each AO as well as the approximate weighting of the different AOs in the Mathematics paper are detailed in **Table 2**.

ASSESSMENT OBJECTIVES (AOs)			APPROX. WEIGHTING
AO1	Knowing	Candidates should be able to: 1. recall and use specific facts, concepts, rules and formulae; 2. classify numbers, expressions and shapes by common properties; 3. read and interpret simple statements or information; 4. perform simple mathematical operations and routine procedures in varied contexts; 5. retrieve information from graphs, tables, texts or other sources; 6. use measuring instruments and choose appropriate units of measurement; 7. solve simple word problems.	55%
AO2	Applying	Candidates should be able to: 1. interpret, use and display data in tables or graphs; 2. form expressions, equations, inequalities to model problem situations; 3. draw geometric figures or diagrams that model problem situations; 4. apply concepts, rules and formulae, appropriate skills, procedures and strategies to solve routine problems.	35 %
AO3	Reasoning	Candidates should be able to: 1. analyse and use relationships to solve problems; 2. make sense of multi-faceted information; 3. develop appropriate strategies to solve non-routine problems in varied contexts; 4. demonstrate logical reasoning in solving routine and non-routine problems; 5. interpret information, make inferences and explain their reasoning with clarity; 6. explain whether results make sense in the context of the problems.	10 %

Table 2: Assessment Objectives and their respective weighting in the Mathematics Paper

5. Paper Description

The Mathematics paper carries a total of 100 marks and is of a duration of 2 hours.

The paper is graded in terms of difficulty across the paper and within sub-questions. Various types of questions are used in this context including Objective-Type questions such as MCQs, Short-Answer Questions and Structured / Long-Answer Questions.

Around 50% of the questions are at basic level, 20% at intermediate level and 30% at proficient level.

Candidates should attempt **all** questions.

5.1 Types of Questions

5.1.1 Multiple-Choice Questions (MCQs)

In the question paper, Multiple-Choice Questions appear as basic level questions consisting of ten items. MCQs may be used to assess any of the AOs defined above. Each MCQ will carry only one mark and four options will be provided with only one correct answer.

5.1.2 1-mark Short-Answer Questions

In general, the 1-mark short-answer questions will assess basic concepts and skills of the *Mathematics TLS Grades 7 to 9* (MIE, 2017).

5.1.3 Short-Answer Questions carrying more than 1 mark

Such questions may comprise one or more parts. Marks will be awarded as follows:

- For questions comprising two or more sub-parts, marks will be awarded for the correct answers, one mark for each part.
- For questions with one part only, 2 or more marks will be awarded for the correct answer. If an incorrect answer is given, intermediate mark(s) may be awarded for a correct method or working shown.

5.1.4 Structured / Long-Answer Questions

For such questions, the candidates will have to show how he/she arrived at a solution (working steps) clearly. Omission of essential working may result in loss of marks.

6. Some other considerations

It is important to highlight a few other important considerations which guided the development of the NCE Mathematics Specimen Paper. They are:

1. The Mathematics paper will be used to measure achievement of candidates from both the Regular and Extended Programmes. Consequently, the paper should provide candidates of **all** abilities with a wide range of opportunities to demonstrate what they know and can do in Mathematics.
2. The *Mathematics Curriculum* is spiral in nature, where learning in one grade builds upon the knowledge, understanding and skills acquired in lower grades.
3. As mentioned earlier, the NCE Assessment will serve different purposes:
 - promotion to grade 10;
 - achievement of Level 2 on the NQF;
 - selection for admission to Academies.

These had important implications for the development of the Mathematics Specimen Paper. They implied that, in line with the established grading structure in **Table 1**, the paper developed should reflect expected outcomes at the end of the NYCBE cycle.

More specifically, to achieve numerical grade 6 or better in the NCE Mathematics Assessment, a candidate needs to score at least 35 marks. However, to achieve these 35 marks and to progress to Grade 10 with confidence, a candidate should be able to demonstrate a minimum level of *Knowing*, *Applying* and *Reasoning* abilities. The paper was developed, therefore, to include accessible, basic tasks under each of the AOs to provide evidence of candidates' acquisition of these competencies.

In parallel, to achieve numerical grade 1 in the NCE Mathematics Assessment, a candidate needs to score at least 85 marks. The paper was also developed, therefore, to provide evidence of the mathematical proficiency attained by high-performing candidates by presenting sufficiently rich and stimulating tasks towards the end of the paper.

With a view to providing a meaningful and positive learning experience to learners, provision has equally been made in the Mathematics paper to:

- include a wide range of different types of questions of varying complexity and requiring different modes of response in the paper.
- minimise the language difficulties which the candidate may encounter. Wherever and whenever possible, the contexts presented or situations given in the paper are described in simple terms.
- encourage the candidate's creativity in solving mathematical problems. Ample working space is given for the candidate to demonstrate his/her mathematical abilities.

7. The Blueprint

The Blueprint provided in this document indicates the topics assessed, the learning objectives and the weightings of the Assessment Objectives in the Specimen Paper.

8. Bibliography

American Educational Research Association (2014). *Standards for educational and psychological testing*.

Legal supplement to the Government Gazette of Mauritius No. 120 of 22 December 2018, *THE EDUCATION ACT – Regulations made by the Minister under section 38 of the Education Act*.

Mauritius Examinations Syndicate (2018). *Annual Programme for the National Certificate of Education (NCE) Assessment for 2020*

Mauritius Institute of Education (2016). *National Curriculum Framework Grades 7, 8 & 9*.

Mauritius Institute of Education (2019). *National Curriculum Framework Syllabus for Grades 7, 8 & 9*.

NCE Mathematics Specimen Paper Blueprint 2020

Content Strand	Topic/Sub-topic	Qu. No	Learning Objectives	Textbook	55	35	10	100
Numbers	Operations	1	Perform arithmetic operations	primary level	1			1
		11(a)	Perform arithmetic operations	primary level	1			1
	Dec/Fractions	4	Convert fractions to decimals	Pg 57 (G7)	1			1
	Fractions	2	Add fractions	Pg 48 (G7)	1			1
		11(c)	Demonstrate an understanding of the concept of fractions	Pg 44 (G7)	1			1
		13	Subtract fractions	Pg 49 (G7)	2			2
	Bodmas	12	Perform operations according to BODMAS	Pg 32 (G7)		1		1
	Percentage	8	Work with percentages	Pg 122 (G7)	1			1
	Indices	3	Recognise and use laws of indices	Pg 134 (G7)	1			1
		14(a)	Use laws of indices	Pg 135 (G7)	2			2
		14(b)	Solve simple equations involving indices	Pg 11 (9)		3		3
	Sequences	11(i)	Complete number patterns of fibonacci sequence	Pg 211 (G9)	1			1
		30(i)	Identify and complete a pattern	pg 216 (G9)		2		2
		30(ii)	Extending a number pattern	pg 216 (G9)			2	2
		30(iii)	Extending and finding the general term	pg 216 (G9)				1
	Ratio	11(g)	Compare two quantities	Pg 126 (G7)	1			1
		11(f)	Solve word problems involving direct proportions	Pg 131 (G7)		1		1
	Decimals	6	Multiply decimals	Pg 147 (G7)	1			1
	Square Roots	9	Find square roots and perfect squares	Pg 27 (G7)	1			1
	Integers	5	Perform arithmetic operations on integers	Pg 20 (G7)	1			1
		11(e)	Use integers in real life situations	Pg 25 (G7)		1		1
Geometry	Angles	18	Finding unknown angle using the notion of corresponding angles, alternate angles and co-interior angles	Pg 79 (G7)				2
	Polygons	11(b)	Identify and name polygon	Pg 233 (G8)	1			1
		11(d)	Identify and use complementary and supplementary angles	Pg 75 (G7)		1		1
	Vectors	21(a)	Express a vector in column form	pg 236 (G9)	2			2
		21(b)	Finding the magnitude of a vector	Pg 236 (G9)		2		2
	Translation	22 (b) (ii)	Finding the translation vector when image and object are given	pg 240 (G9)		2		2
	Coordinates	22(a)	Finding the gradient /equation of line of an inclined line	pg 66 (G9)	3			3
	Trigonometry	27	Solve trigonometric problems in 2-D to find unknown sides	pg 135 (G9)			4	4
	Reflection	22(b) (i)	Reflect points	Pg 232 (G7)		1		1
	Basic Constructions	19	Construct triangles given 3 sides	Pg 262 (G8)		2		2

Measurements	Speed	16	Finding speed given time and distance	Pg 157 (G7)	2													2	14
	Money	17	Convert from one currency to another	Pg 143 (G7)						2								2	
	Mass/Length/Time	7	Convert units of time and length	Pg 97 / pg 147 (G7)	2													2	
	TSA / volume of right prisms	31(a)	Find the volume of a cylinder	pg 171 (G9)					2									2	
Algebra		31(b)(i)	Solve problems involving areas	Pg 187 (G9)												3		6	29
		12(b)	adding alg. terms	Pg 175 (G7)	1													1	
	Algebraic exp/terms	29(i)	Expanding a binomial	pg 16 (G9)				2										2	
	LCM/HCF of alg exp	20(b)	Finding the LCM of algebraic expressions	Pg 110 (G8)						2								2	
	Operations on Algebra	23(b)	Changing the subject of the formula	Pg 41 (G9)							2							2	
	Simultaneous Equations	26	Solve simultaneous equations	Pg 63 (G9)				4										4	
	Matrices	11(h)	Finding the order of a matrix	Pg 142 (G9)	1													1	
		28(a)	Solve matrix equations	Pg 143 (G9)					2									2	
		28(b)	Multiply matrices	Pg 152 (G9)					3									3	
	Sets	11(i)	Finding the cardinal number of a set	Pg 251 (G7)	1													1	
	Factorisation	20(a)	Factorise algebraic expressions	Pg 179 (G7)	1													1	
	Evaluation	23(a)	Evaluate alg. Exp. By substituting numbers	Pg 180 (G7)	2													2	
	Inequalities	15(a)	Solve simple linear inequalities	Pg 171 (G8)	2													2	
		15(b)	Demonstrate an understanding of linear inequalities	Pg 176 (G8)						1								1	
	Quadratics	29(ii)	Formulate quadratic equations	Pg 35 (G9)														2	
		29(iii)	Solving quadratic equations	Pg 36 (G9)					3									3	
	Rep of data	10	Interpret a pie chart	Pg 259 (G9)	1													1	
Statistics	Mean/Mode/ Median	24(a) (i)	Calculating mean of discrete set of data	Pg 260 (G9)							2							2	6
		24 (a) (ii)	Finding the mode of a discrete set of data	Pg 266 (G9)				1										1	
		24(b)	Demonstrate an understanding of median	Pg 265 (G9)												2		2	
Probability	Simple Probability	25(a)(i)	Finding probabilities of simple events	pg 275 (G9)												1		1	4
		25(a) (ii)	Finding probabilities of simple events	pg 275 (G9)							1							1	
		25(b)	Finding probabilities of simple events	pg 275 (G9)												2		2	
Total					32	10	15			18	5	9				5	6	100	100
Grand Total						57				32						11			

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Index Number:

NATIONAL CERTIFICATE OF EDUCATION

Specimen paper for first assessment in October 2020

MATHEMATICS (N 510)

TIME: 2 HOURS

Additional Material: Geometrical Instruments

READ THESE INSTRUCTIONS FIRST

1. Write your index number in the space provided above.
2. Write in dark blue or black ink.
You may use an HB pencil for any diagrams. Do not use correction fluid.
3. All answers should be written in the spaces provided in the question paper.
Diagrams are not drawn to scale unless otherwise specified.
3. Answer **ALL** questions.
4. All necessary workings should be shown in the spaces provided.
5. Check that this assessment paper consists of 31 questions printed on 29 pages from page 2 to page 30.
6. **ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.**
7. The total number of marks for this paper is **100**.
The number of marks is given in brackets [] at the end of each question or part question.

1. Work out

$$\begin{array}{r} 324 \\ + 183 \\ \hline \\ \hline \end{array}$$

Answer: [1]

2. Evaluate $\frac{2}{5} + \frac{1}{5}$

Answer: [1]

3. Simplify $a^3 \times a^4$

Answer: [1]

4. Express $\frac{7}{10}$ as a decimal

Answer: [1]

5. Evaluate $4 - 7$

Answer: [1]

6. Calculate 1.3×4

Answer: [1]

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7. (a) 6 minutes = seconds

(b) 5700 m = km.

[2]

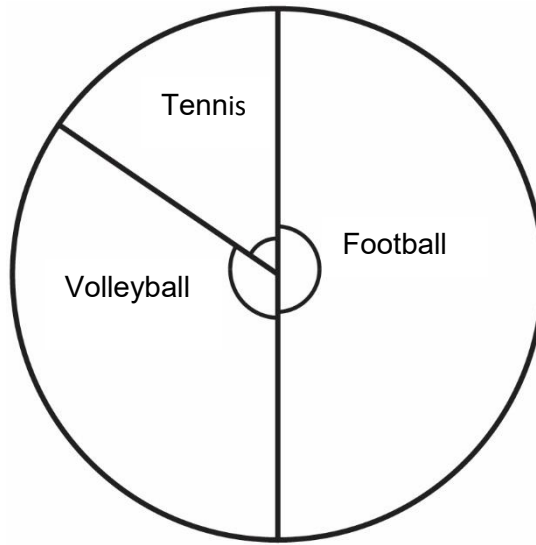
8. Evaluate 20% of 120

Answer: [1]

9. Find $\sqrt{144}$

Answer: [1]

10. The pie chart below shows the different sports students like in a class.



Which sport do students like **most**?

Answer: [1]

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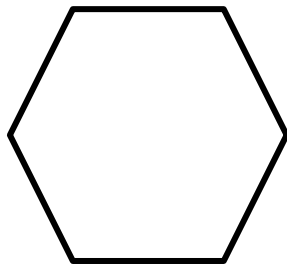
11. Circle the letter corresponding to the correct answer in each of the following items.

(a) $1284 \div 6 =$

- A 212
- B 213
- C 214
- D 215

[1]

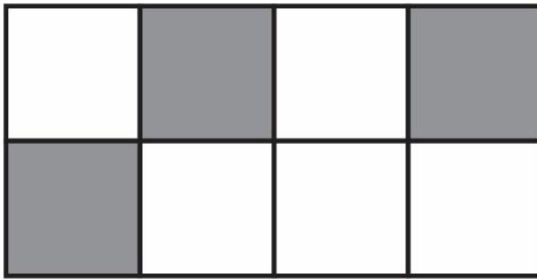
- (b) What is the name given to the polygon shown below?



- A Hexagon
- B Decagon
- C Nonagon
- D Pentagon

[1]

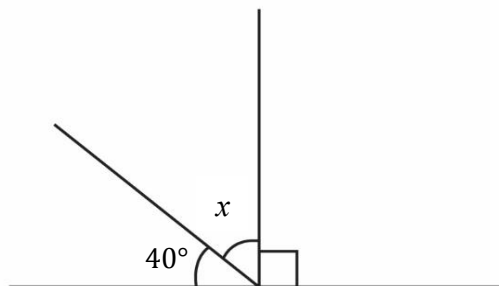
- (c) What fraction of the diagram is **shaded**?



- A** $\frac{2}{8}$
B $\frac{3}{8}$
C $\frac{4}{8}$
D $\frac{5}{8}$

[1]

- (d) What is the value of angle x ?

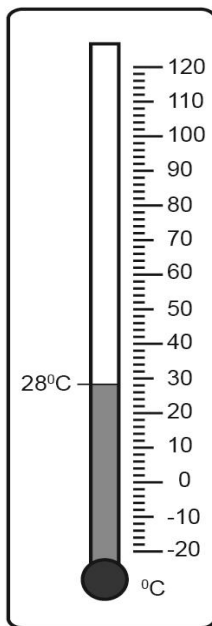


- A** 40°
B 50°
C 130°
D 140°

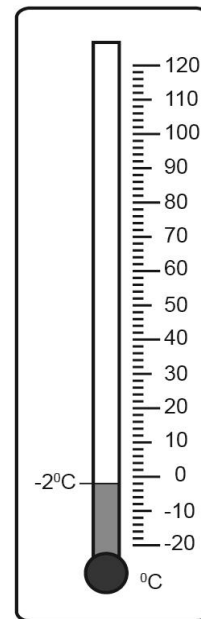
[1]

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- (e) Observe the temperatures indicated on the two thermometers below carefully.



Temperature in Port Louis



Temperature in London

What is the **difference** between the temperatures in Port Louis and London?

- A** 26°C **B** 28°C **C** 30°C **D** 32°C

[1]

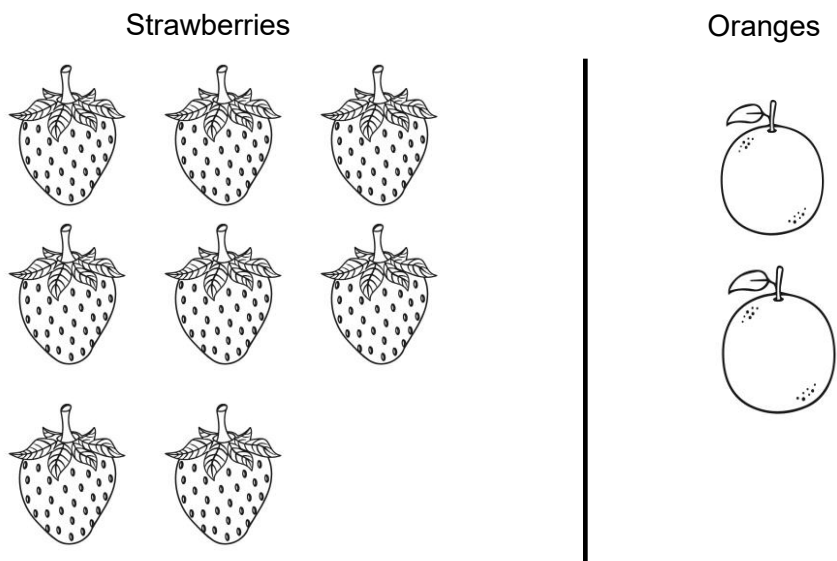
- (f) The cost of 7 pencils is Rs 35.

What is the cost of 1 pencil?

- A** Rs 5
B Rs 28
C Rs 42
D Rs 245

[1]

- (g) What ratio, **expressed in its simplest form**, represents the number of strawberries to the number of oranges?



- A** 4 : 1
B 1 : 4
C 8 : 2
D 2 : 8

[1]

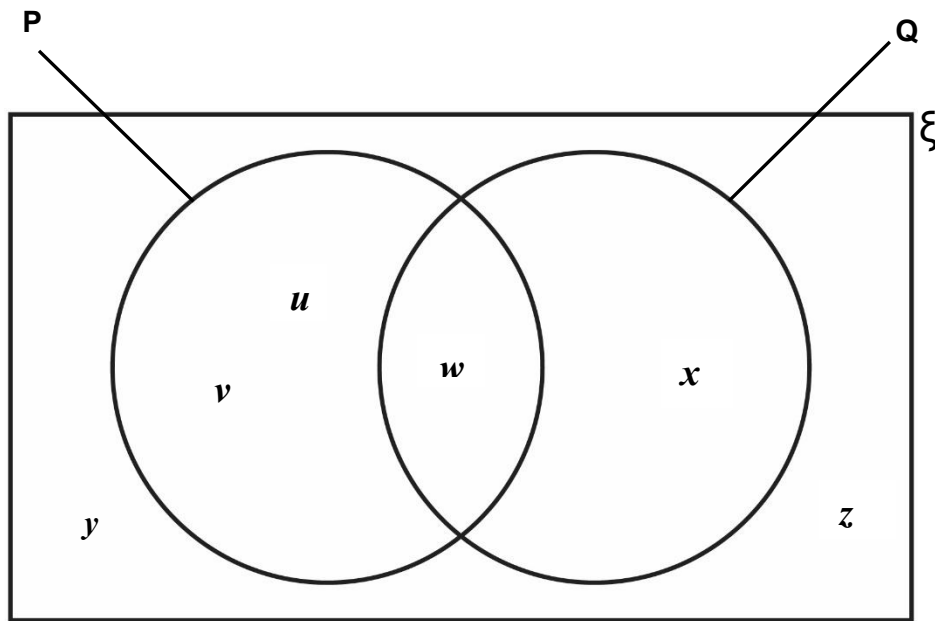
- (h) Given that matrix $P = \begin{pmatrix} 1 & 5 & 6 \\ -2 & 3 & 0 \end{pmatrix}$, what is the **order** of matrix P?

- A** 3×2
B 3×3
C 2×2
D 2×3

[1]

Please turn over this page

- (i) Study the Venn diagram below carefully.



What is the value of $n(P \cup Q)$?

- A** 1
B 4
C 5
D 6

[1]

- (j) Observe the **Fibonacci sequence** given below.

2, 3, 5, 8, ...

What is the next term of the sequence?

- A** 10
B 11
C 12
D 13

[1]

12 (a) Evaluate $20 - 7 \times 2$

Answer: [1]

(b) Simplify $3x + 4x$

Answer: [1]

13. Calculate $5\frac{2}{3} - 2\frac{1}{6}$, giving your answer in its simplest form.

Answer: [2]

14. (a) Evaluate $(5^4)^{\frac{1}{2}}$

Answer: [2]

(b) Given that $2^{x+1} = 32$, find the value of x .

Answer: $x = \dots\dots\dots$ [3]

15. (a) Solve the inequality $2x - 5 < 17$.

Answer: $\dots\dots\dots$ [2]

(b) Hence, write down the **greatest** possible integer value of x .

Answer: $\dots\dots\dots$ [1]

16. Lea travels a distance of 147 kilometres in 3 hours.
Find her average speed in kilometres per hour.

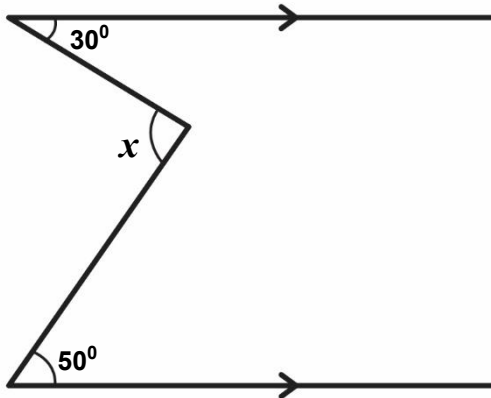
Answer: km/h [2]

-
17. Sanju goes to the bank to exchange Rs 1400 into US \$.
How many US \$ does he get, if US \$ 1 = Rs 35?

Answer: US \$..... [2]

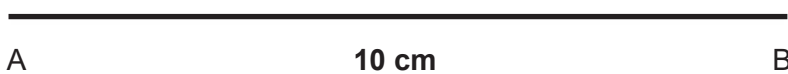
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18. Calculate angle x .



Answer: $x = \dots\dots\dots$ [2]

19. Using a set of compasses, **construct** triangle ABC ,
where $AB = 10 \text{ cm}$, $AC = 7 \text{ cm}$ and $BC = 8 \text{ cm}$.
The line AB has been drawn for you.



[2]

20. (a) Factorise $5x - 10$.

Answer: [1]

- (b) Find the LCM of $2x^2y$ and $4xy^2$.

Answer: [2]

-
21. Given $\mathbf{p} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$,

Find (a) $2\mathbf{p} + \mathbf{q}$,

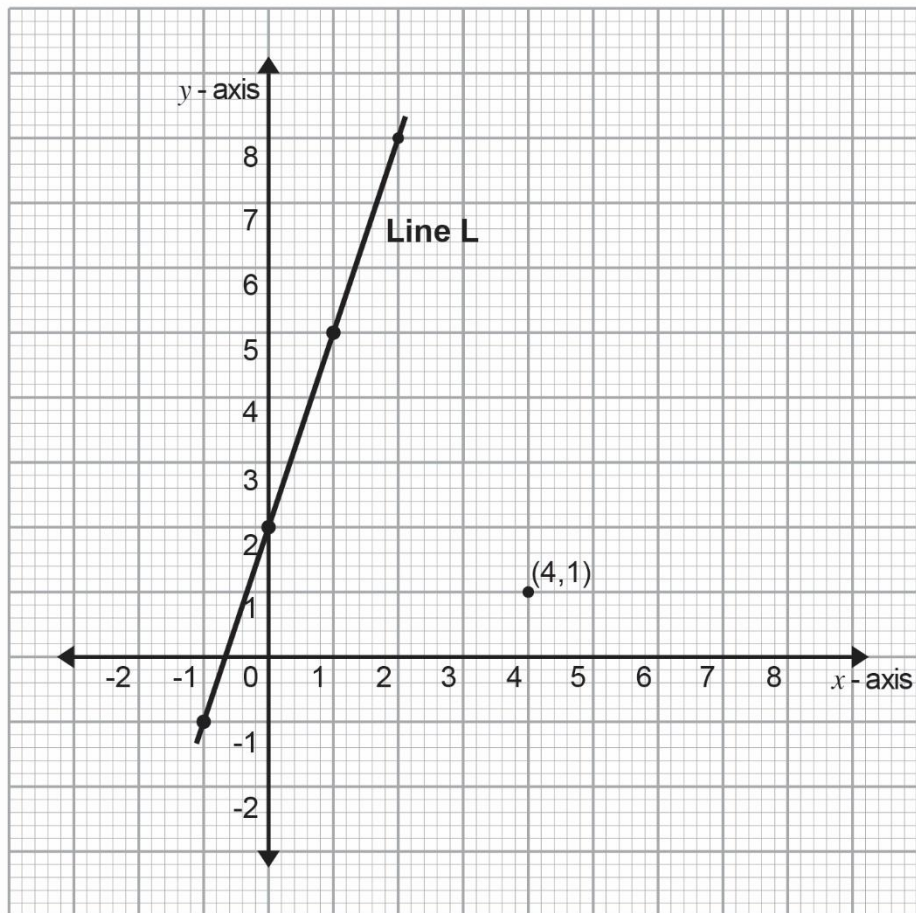
Answer: [2]

- (b) $|\mathbf{p}|$.

Answer: [2]

Please turn over this page

22. The graph below shows a straight **Line L**.



(a) (i) Find the gradient of **Line L**.

Answer: [2]

(ii) Write down the equation of the **Line L**.

Answer: [1]

- (b) (i) Write down the coordinates of the image of the point $(4,1)$ under a reflection in the x -axis.

Answer: [1]

- (ii) Find the translation vector when the point $(3,2)$ is mapped onto $(1,3)$.

Answer: [2]

23. Given that $y = 3u - v$,

(a) find the value of y when $u = 5$ and $v = 1$.

Answer: [2]

(b) Express u in terms of v and y .

Answer: [2]

- 24** (a) The following numbers represent the marks obtained by six students in a class test.

5 3 2 3 8 3

- (i) Find the mean mark.

Answer:..... [2]

- (ii) Find the mode.

Answer:..... [1]

- (b) The following numbers are arranged in ascending order.

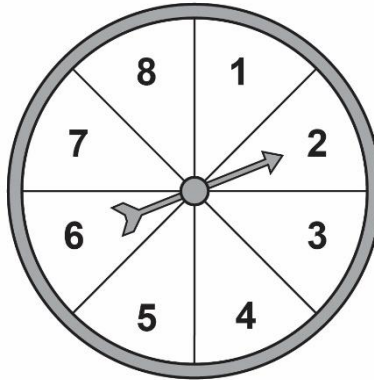
1, 4, 6, 6, x , 7, 7, 10

If the median of the numbers is $6\frac{1}{2}$, find the value of x .

Answer: $x =$ [2]

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25. (a) The spinner below is rotated once.



Find the probability that the arrow will point to

- (i) number 9,

Answer:..... [1]

- (ii) a number greater than 6.

Answer:..... [1]

- (b) The table below shows the mode of transport used by 10 workers to go to work.

Mode of Transport	Number of workers
Bus	5
Motorcycle	2
Car	3

A worker is chosen at random.

- (i) Find the probability that the worker travels by **bus**.

Answer:..... [1]

- (ii) Hence, draw an arrow on the probability scale to show the probability that the worker chosen travels by **bus**.



[1]

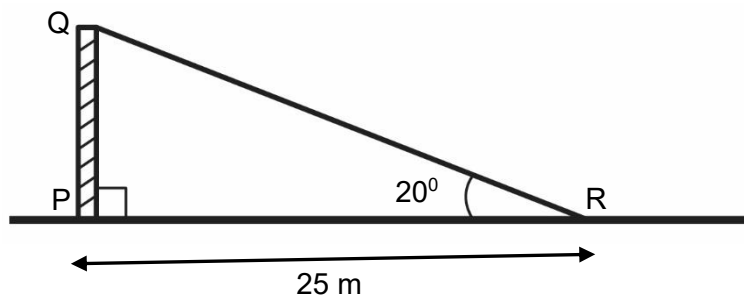
26. Solve the simultaneous equations,

$$2x + 3y = 11$$

$$x - 2y = 2$$

Answer: $x = \dots\dots\dots$ $y = \dots\dots\dots$ [4]

27. The diagram below shows a vertical pole PQ. The points P and R are on level ground. R is 25 metres from the foot P of the pole and angle PRQ = 20° .



Using the information given below,

[$\sin 20^\circ = 0.342$, $\cos 20^\circ = 0.940$, $\tan 20^\circ = 0.364$]

Estimate the height of the pole PQ, giving your answer to the nearest whole number.

Answer:.....m [4]

28. $\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 5 & 2 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 2 & 2 \\ 4 & 1 \end{pmatrix}$ $\mathbf{C} = \begin{pmatrix} 3 & -1 \\ 5 & 3-y \end{pmatrix}$

(a) If $\mathbf{A} = \mathbf{C}$, find the value of y .

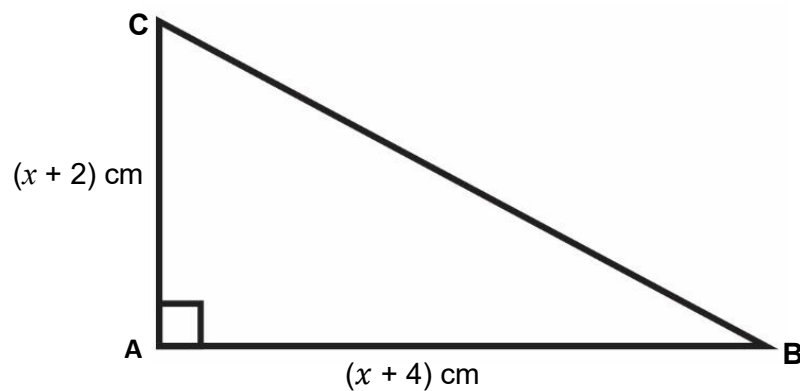
Answer: $y = \dots\dots\dots$ [2]

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(b) Evaluate **AB**

Answer:..... [3]

29. In triangle ABC, $AB = (x + 4)$ cm, $AC = (x + 2)$ cm and angle BAC is right-angled.



(i) Expand $(x + 4)(x + 2)$

Answer:..... [2]

- (ii) The area of triangle ABC is 24 cm^2 . Use your answer in part (i) to show that $x^2 + 6x - 40 = 0$.

[2]

- (iii) Solve the equation $x^2 + 6x - 40 = 0$ to find the values of x .

Answer: $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

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30. Observe the pattern below.

Figure 1



Figure 2

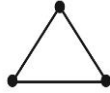


Figure 3

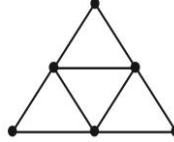
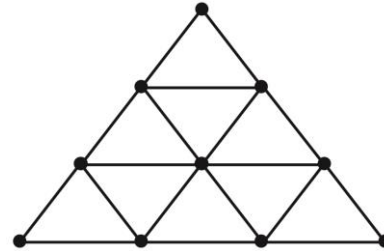


Figure 4



(i) Write the **sequence** for the number of **dots** in the above figures.

Answer:,,, [2]

(ii) Write down the next two terms in the sequence obtained in part (i).

Answer:, [2]

- (iii) Showing your working clearly, determine which one of the following expressions, **(A) or (B)**, gives the n^{th} term of the sequence..

(A): $n^{\text{th}} \text{ term} = \frac{n(n-1)}{2}$

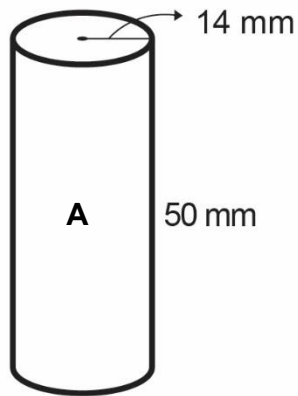
Or

(B): $n^{\text{th}} \text{ term} = \frac{n(n+1)}{2}$

Answer: [1]

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31. (a) Solid metal **cylinder A** has radius 14 mm and height 50 mm.

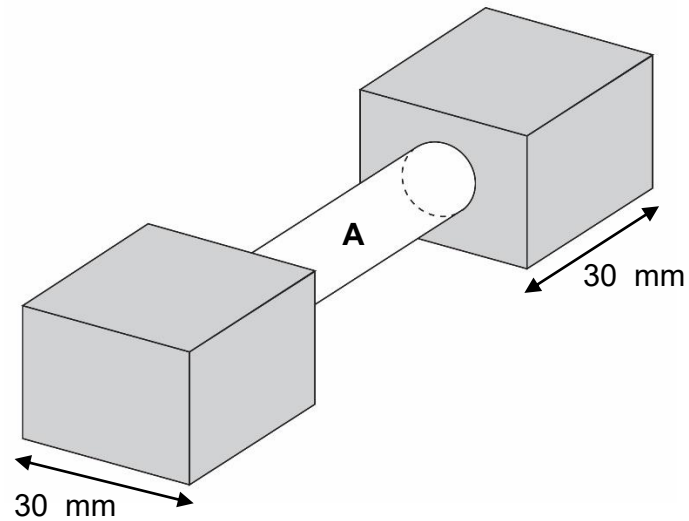


Taking $\pi = \frac{22}{7}$, calculate its volume.

Answer:..... mm³ [2]

- (b) An apparatus is made by fixing 2 **identical** metal cubes to cylinder **A** as shown in the diagram below.

The length of each edge of the metal cube is 30 mm.



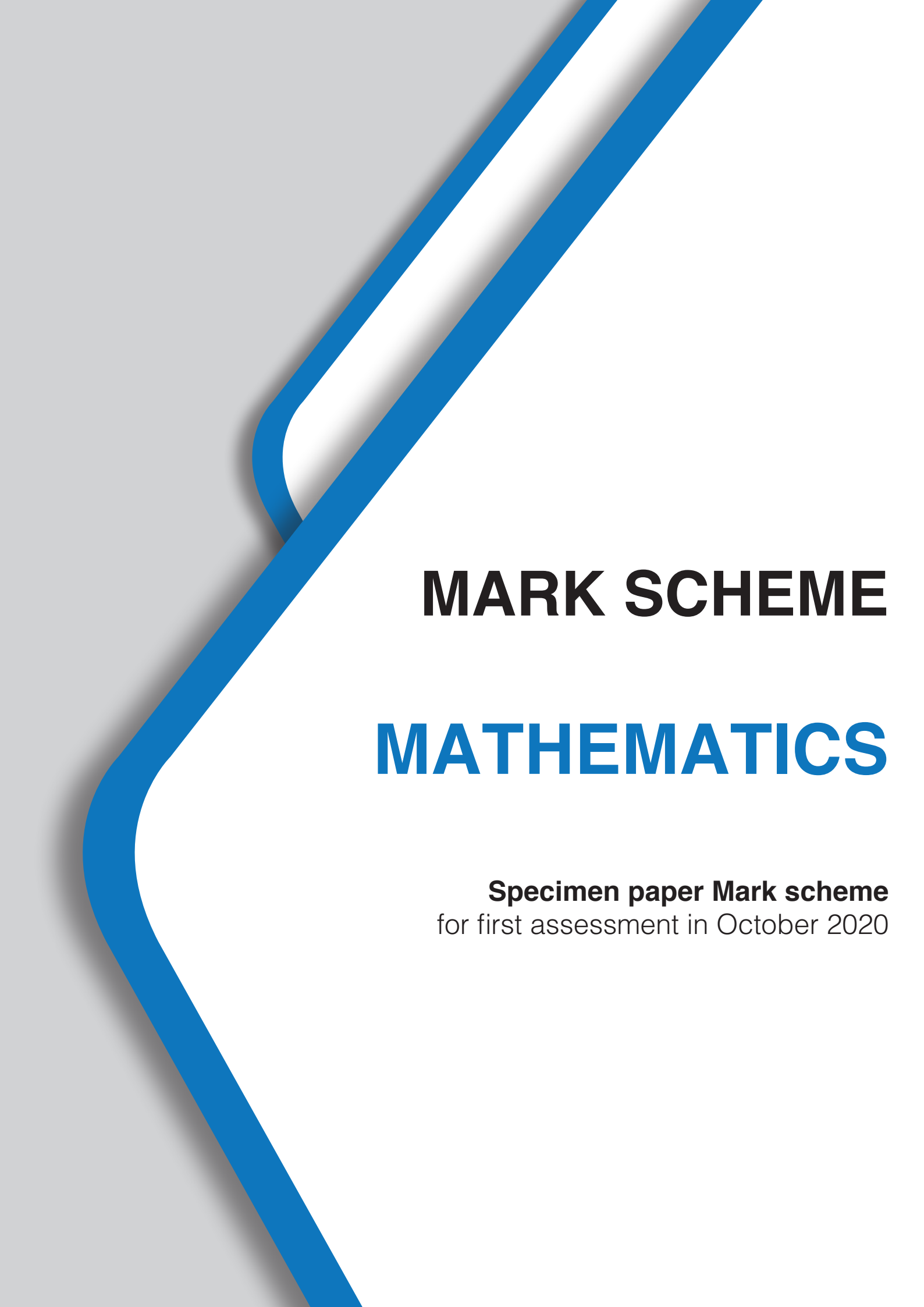
Calculate the **total surface area** of the apparatus.

Answer: mm² **[6]**

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MARK SCHEME

MATHEMATICS

Specimen paper Mark scheme
for first assessment in October 2020

The NCE Mathematics Specimen Mark scheme

The Mathematics Specimen Mark Scheme serves to indicate the principles of allocation of marks. It provides the reader with an insight about why, when and how to reward a candidate's achievement in Mathematics.

Any proposed mark scheme is generally not a finalised document and can be adjusted according to candidates' responses.

The methods indicated in the current mark scheme is not exhaustive.

Generic Marking principles

These Generic Marking principles must be applied by all examiners when marking candidates' answers, alongside the specific content of the mark scheme for each question.

- Marks awarded are always **whole marks** (Do not award half marks or fractions).
- Marks awarded must be in line with :
 1. the specific content of the mark scheme
 2. the specific skills defined in the mark scheme
 3. the standard of the response required by the candidate
- Marks must be awarded **positively** as follows
 1. Marks are awarded for correct/valid answers as defined in the mark scheme. However credit is given for valid answers which go beyond the scope of the syllabus.
 2. Marks are awarded for what candidates know and can do.
- Rules must be applied and instructions followed consistently by markers.

Important notes about the mark scheme

B marks are independent marks, which do not depend on any other marks. For a B mark to be scored, the point to which it refers must actually be seen in the candidate's answer.

M marks are method marks upon which accuracy marks (A marks) later depend. For an M mark to be scored, the point to which it refers must be seen in a candidate's answer. If a candidate fails to score a particular M mark, then none of the dependent A marks can be scored.

A marks are accuracy or answer marks which either depend on an M mark, or which are one of the ways which allow a C mark to be scored.

C marks are compensatory method marks, which can be scored even if the points to which they refer, are not written down by the candidate, provided subsequent working gives evidence that they must have known it.

Abbreviations

c.a.o. means "correct answer only"

soi means seen or implied

ecf means error carried forward

oe means "or equivalent"

asc means all signs changed

FT means "Follow through". This indicates that if a candidate has made an earlier mistake and has carried his incorrect value forward to subsequent stages of working, he may be given marks indicated by FT provided his subsequent working is correct, bearing in mind his earlier mistake. This prevents a candidate being penalised more than once for a particular mistake, but only applies to marks annotated "e.c.f."

Question	Answers	Mark	Remarks
1	507	B1	
2	$\frac{3}{5}$	B1	
3	a^7	B1	
4	0.7	B1	Accept .7 Ignore superfluous zeros
5	-3	B1	
6	5.2 or $5\frac{1}{5}$	B1	
7(a)	360 seconds	B1	
7 (b)	5.7 km or $5\frac{7}{10}$	B1	
8	24	B1	
9	12	B1	Accept ± 12 or -12
10	Football	B1	
11 (a)	C	B1	
11(b)	A	B1	
11(c)	B	B1	
11(d)	B	B1	
11(e)	C	B1	
11(f)	A	B1	
11(g)	A	B1	
11(h)	D	B1	
11(i)	B	B1	
11(j)	D	B1	

12(a)	6	B1	
12(b)	$7x$	B1	
13	$3\frac{1}{2}$ or 3.5 or $\frac{7}{2}$	B2	After B0, B1 for $\frac{17}{3} - \frac{13}{6}$ seen or for $5\frac{4}{6} - 2\frac{1}{6}$ seen or for $3\frac{3}{6}$ seen
14(a)	25	B2	After B0, B1 for 5^2 seen
(b)	4	B3	B1 for 2^5 soi M1 for $x + 1 = 5$ soi A1 for $x = 4$ or <u>Alternative Method</u> M1 for $2^x = \frac{32}{2}$ or $2^x = 16$ seen M1 for $2^x = 2^4$ A1 for $x = 4$
15(a)	$x < 11$	B2	After B0, B1 for $2x < 22$ or After B0, B1 for $x < \frac{\text{their } 22}{2}$ correctly evaluated
15(b)	10	A1	FT on their (a) provided inequality is in the form $x < k$

16	49	B2	After B0 M1 for $\frac{147}{3}$ seen
17	40	B2	After B0 M1 for $\frac{1400}{35}$ oe
18	80^0	B2	After B0 B1 for $30^0 + 50^0$ seen or B1 for 130^0 or 150^0 seen
19	The two lines drawn correctly. Accept $\pm 1mm$ for length	B2	After B0, M1 for one line drawn correctly
20 (a)	$5(x-2)$	B1	Accept $(x-2) 5$
(b)	$4x^2y^2$	B2	Accept answer given in expanded form After B0, B1 for either 4 or x^2y^2 seen
21(a)	$\begin{pmatrix} 5 \\ 13 \end{pmatrix}$	B2	After B0, B1 for 1 correct component or B1 for $\begin{pmatrix} 5 \\ 13 \end{pmatrix}$ or $\frac{5}{13}$
21(b)	5	B2	After B0, M1 for $\sqrt{3^2 + 4^2}$ or $\sqrt{9 + 16}$ or $\sqrt{25}$

22(a) (i)	3	B2	After B0, M1 for either using the formula $\frac{y_2 - y_1}{x_2 - x_1} / \frac{y_1 - y_2}{x_1 - x_2}$ or for using $\frac{\text{change in } y \text{ values}}{\text{change in } x \text{ values}}$ directly from the graph
22(a)(ii)	$y = 3x + 2$	B1	
22 (b) (i)	$(4, -1)$	B1	
22(b) (ii)	$\begin{pmatrix} -2 \\ 1 \end{pmatrix}$	B2	After B0, B1 for one correct component or B1 for $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ or $(-2, 1)$ or $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$
23(a)	14	B2	After B0, B1 for $3(5) - 1$ or B1 for $15 - 1$ or B1 for $(\text{their } 15) - 1$ correctly evaluated
23(b)	$u = \frac{y+v}{3}$ or equivalent	B2	After B0, M1 for $3u = y + v$ asc
24 (a) (i)	4	B2	After B0 M1 for $\frac{5+3+2+3+8+3}{6}$
24 (a) (ii)	3	B1	

24(b)	7	B2	After B0, M1 for $\frac{x+6}{2} = 6\frac{1}{2}$ or M1 for $x + 6 = 13$
25(a) (i)	0	B1	Accept $\frac{0}{8}$
(a) (ii)	$\frac{1}{4}$	B1	Accept $\frac{2}{8}$
25 (b)(i)	$\frac{1}{2}$ or $\frac{5}{10}$ or 0.5	B1	
25 (b) (ii)	Correct placement of arrow	B1	
26	Correct method to eliminate one variable reaching $ax = b$ or $cy = d$	M2	M1 for $4x + 6y = 22$ or M1 for $3x - 6y = 6$ or M1 for $2x - 4y = 4$ or M1 for $x = 2 + 2y$
	$x = 4$ $y = 1$	A2	A1 for either $x = 4$ or $y = 1$ or after A0, C1 for a pair of values which satisfy either equation or for correct answers with no working
27	9	M1	M1 for $\tan 20^\circ = \frac{PQ}{25}$
		M1	M1 for $PQ = 25 \times \tan 20^\circ$
		A1	A1 for 9.1

		B1	B1 for 9 FT from (their 9.1)
28(a)	1	B2	B0, M1 for $3 - y = 2$
28(b)	$\begin{pmatrix} 2 & 5 \\ 18 & 12 \end{pmatrix}$	B3	B2 for any three correct entries B1 for any two correct entries
29 (i)	$x^2 + 6x + 8$	B2	After B0, M1 for any two correct terms in a three term expression or M1 for $x^2 + 2x$ or M1 for $4x + 8$
29 (ii)	$x^2 + 6x - 40 = 0$ correctly found	B2	After B0, M1 for $\frac{1}{2}(x + 4)(x + 2) = 24$ oe
29(iii)	$\left. \begin{array}{l} (x + 10)(x - 4) \\ \textbf{or} \\ (x - 10)(x + 4) \end{array} \right] \\ x = -10 \\ x = 4$	M1 A1 A1	
30(i)	1,3,6,10	A2	After A0, B1 for any three correct answers and in correct order
30(ii)	15 21	B1 B1	

30(iii)	B correctly shown	B1	
31(a)	30 800	B2	After B0 M1 for calculating volume of cylinder that is $\frac{22}{7} \times 14 \times 14 \times 50$
31(b)	13 968	B1 B1 B1 B1 M1 A1	B1 for calculating area if square : 30×30 oe B1 for calculating area of circle: $\frac{22}{7} \times 14 \times 14$ oe B1 for calculating curved surface area: $2 \times \frac{22}{7} \times 14 \times 50$ B1 for $10(30 \times 30)$ or $12(30 \times 30)$ or $2(\frac{22}{7} \times 14 \times 14)$ seen M1 for $10 \times \text{their } (30 \times 30)$ $+ 2 \times \text{their } [(30 \times 30) - (\frac{22}{7} \times 14 \times 14)]$ $+ \text{their } (2 \times \frac{22}{7} \times 14 \times 50)$ or $12 \times (\text{their } 30 \times 30)$ $- 2 \times \text{their } (\frac{22}{7} \times 14 \times 14)$ $+ \text{their } (2 \times \frac{22}{7} \times 14 \times 50)$ 13968